



Exploring Clustering Methods in Machine Learning

In machine learning, clustering is a technique that groups data points. Ideally, data points in a cluster should have similarities in properties or features, which differ from those not in the same cluster. This article discusses two clustering algorithms – k-means and DBSCAN.



One of the most popular techniques in data science is clustering, a machine learning (ML) technique for identifying similar groups of data in a data set. Entities within each group share comparatively more similarities with each other compared to with those from other groups. Clustering means finding clusters in an unsupervised data set. A cluster is a group of data points or objects in a data set that are similar to other objects in the group and dissimilar to data points in other clusters. Clustering is useful in several exploratory pattern-analysis, grouping, decision-making and machine-learning situations, including data mining, document

retrieval, image segmentation and pattern classification. However, generally, the data is unlabelled and the process is unsupervised in clustering.

For example, we can use a clustering algorithm such as k-means to group similar customers as mentioned and assign them to a cluster, based on whether they share similar attributes, such as age, education and so on.

Some of the most popular applications of clustering include market segmentation, medical imaging, image segmentation, anomaly detection, social network analysis and recommendation engines. In banking, analysts examine clusters of normal transactions to find the patterns of fraudulent credit